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Domain-adaptive faster R-CNN for non-PPE identification on construction sites from body-worn and general images

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Ensuring consistent compliance with Personal Protective Equipment (PPE) requirements on construction sites is crucial for worker safety. Although deep learning–based methods already perform well in detecting non-PPE cases, there is still scope to further improve accuracy. Progress is hindered by the difficulty of building representative datasets: strict regulations mandate PPE usage, so genuine non-PPE instances are rare, even though such examples are essential for training robust detectors. To address this challenge, this study develops a Domain Adaptation (DA)–based Faster Region-Based Convolutional Neural Network (Faster R-CNN) for detecting five non-PPE categories: "Non-helmet", "Non-mask", "Non-glove", "Non-vest", and "Non-shoes". The proposed framework augments a standard Faster R-CNN with image-level and instance-level adversarial domain classifiers connected through gradient reversal layers, enabling the model to learn domain-invariant features while preserving detection accuracy. The approach leverages a fully labeled construction-site dataset alongside a general-context dataset, allowing the detector to exploit abundant surrogate non-PPE examples and alleviate the scarcity of real on-site violations. Among the evaluated backbones, ResNet-152 combined with comprehensive data augmentation and tuned hyperparameters achieved the best performance, reaching an mean Average Precision (mAP) of 86.84% on previously unseen construction-site images. Overall, the DA-enhanced detector outperformed conventional supervised baselines by up to 14 mAP points. These findings indicate that combining DA with systematic data augmentation improves the robustness of non-PPE detection under realistic construction-site conditions and provides a practical foundation for extending the approach to broader safety-monitoring applications.

Keywords Personal protective equipment, Site monitoring, Deep learning, Domain adaptation, Faster region-based convolutional neural network

Personal Protective Equipment (PPE), including helmets, vests, and other protective apparel, is essential on construction sites for preventing occupational accidents. Nonetheless, compliance frequently suffers as workers often neglect PPE due to discomfort or limited safety awareness[1]. Ensuring consistent PPE use among all site personnel is therefore critical for mitigating accident risks. Traditionally, PPE compliance monitoring relies on manually examining video surveillance or conducting physical site inspections, both of which are resource-intensive and time-consuming tasks. Additionally, inspectors reviewing video footage may unintentionally overlook violations due to fatigue caused by prolonged monitoring[2–4].

Deep learning-based object detection techniques offer an effective automated approach for detecting non-use of PPE through images captured by surveillance or body-mounted cameras. These methods utilize bounding boxes to pinpoint and categorize specific objects within images. By leveraging large, annotated datasets collected under varying conditions—including different lighting scenarios and object scales—deep learning models can significantly streamline and enhance the accuracy of the inspection process[5,6]. However, creating extensive datasets that accurately represent real-world construction site scenarios without PPE remains difficult because strict safety regulations reduce the occurrence of genuine PPE non-compliance cases.

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To address the scarcity of non-PPE examples, one practical approach is to incorporate general-context source datasets—such as images from public streets or everyday environments—where people typically do not wear PPE. In domain adaptation, these general-context images constitute the source domain, which is typically easier to acquire or more abundant[7,8]. On the other hand, the real construction-site images form the target domain, representing the actual environment in which the model needs to perform non-PPE detection. Figure 1 illustrates this shift: construction-site images often cluster together in one region of the feature space, while general-context images form their own cluster in another region. Despite some overlap—likely because both sets depict people—noticeable differences emerge in attire, background, and camera perspectives. Such discrepancies can diminish model accuracy if most training data come from general-context images that do not fully reflect real construction-site conditions.

When integrated with deep learning, Domain Adaptation (DA) addresses the domain shift problem by aligning the source and target domain distributions[9]. By embedding target data into a shared feature space alongside source data, DA substantially reduces this mismatch. Such alignment mitigates the performance drop typically observed when a model trained on one distribution is deployed in a markedly different one, while still leveraging the limited imagery available from the target region[10,11]. Although existing studies have shown that deep learning–based techniques can reliably detect non-PPE scenarios, there is still room for further accuracy improvements. Consequently, DA outperforms standard deep learning by enabling general-context–trained models to adapt effectively—even when construction-site, non-PPE data remain scarce.

In this paper, a deep learning framework based on DA–Faster Region-Based Convolutional Neural Network (Faster R-CNN) is introduced to improve the reliability of non-PPE detection on construction sites. The approach leverages a fully labeled construction-site dataset together with a fully labeled general-context dataset. The main contributions are as follows:

1) The proposed framework detects five critical PPE categories—helmet, mask, glove, vest, and shoes—even when real construction-site images are not available for training.

2) The experimental analysis identifies the top-performing model for each architecture–dataset configuration; the DA-Faster R-CNN with a ResNet-152 backbone trained on augmented source and target data achieves a mean Average Precision (mAP) of 86.84% on the construction-site test set at approximately 33 frames per second.

3) When the proposed DA framework was applied, detection performance increased by up to 14 mAP points compared with supervised Faster R-CNN and by up to 39 mAP points compared with the baseline method, using the same test images.

Literature review

Supervised learning

In recent decades, deep learning—especially Convolutional Neural Networks (CNNs)—has extensively been employed for non-PPE detection tasks. Fang et al.[12], for example, utilized Faster R-CNN for detecting the absence of safety helmets, achieving accuracies ranging between 90.1% and 98.4% across various scenarios. Gu et al.[13] improved Faster R-CNN through multi-scale training, advanced anchor strategies, and online hard example mining, enhancing helmet detection accuracy by approximately 7% compared to the original method. Yang et al.[14] implemented an image pyramid structure combined with You Only Look Once version 3 (YOLOv3), resulting in an accuracy of 95.13% with an inference time of just 0.017 s per image—outperforming Single Shot MultiBox Detector (SSD) and Faster R-CNN.

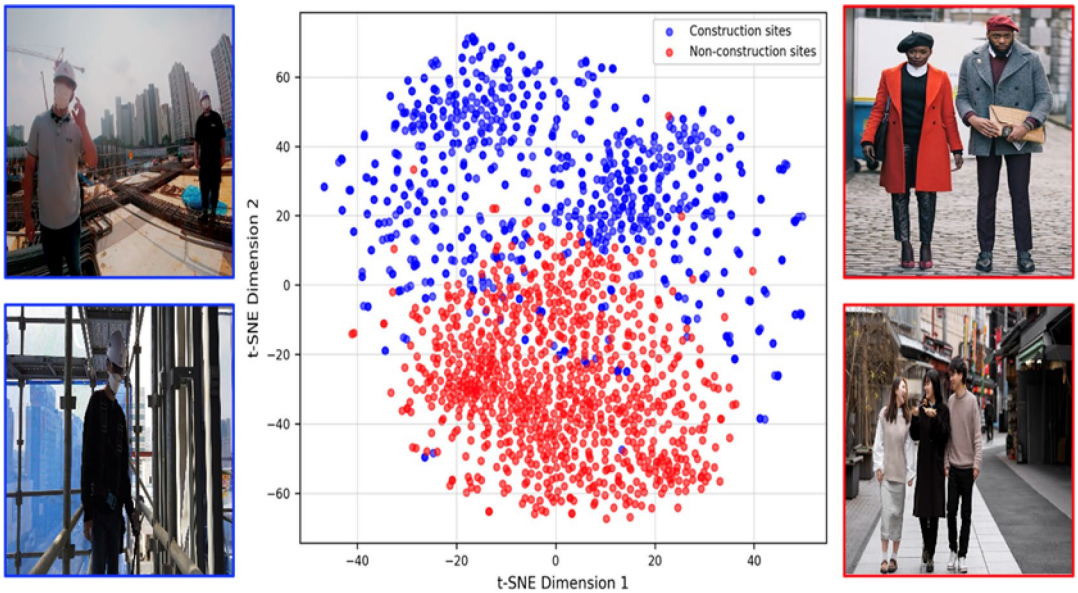


Fig. 1. Visualization of domain shift in two datasets.

Yan and Wang[15] modified YOLOv3's backbone by integrating densely connected convolutional networks, significantly improving detection speed by 52% and accuracy by 5.7% without increasing inference time. Shen et al.[16] introduced a unique approach by employing a single-stage face detector to recognize safety helmets and subsequently applying a regression model linking faces to helmets. Nath[17] combined YOLOv3 with Bayesian inference to identify helmets and vests, achieving a detection accuracy of 71.23%. Wang et al.[18] compared YOLOv5 variants and found YOLOv5x to have the highest accuracy (86.55%), whereas YOLOv5s was the fastest (0.019 s per frame). Lee et al.[19] presented a real-time detection framework based on You Only Look At CoefficienTs (YOLACT), attaining 66.4% accuracy in helmet and vest detection. Nguyen et al.[20] utilized YOLOv5 and YOLOv8, enhanced with a seahorse optimization algorithm, where YOLOv5 demonstrated an average detection accuracy of 66% across multiple PPE categories (helmet, mask, glove, vest, shoes).

Although deep learning methods effectively detect non-PPE usage, their performance heavily depends on extensive datasets accurately reflecting real-world scenarios. However, stringent safety regulations result in limited availability of authentic PPE violation imagery. Consequently, datasets from general contexts—such as street or office environments where PPE usage isn't expected—serve as substitutes. Nevertheless, these datasets often fail to replicate actual construction environments accurately, creating a domain shift problem: models trained on non-construction images tend to generalize poorly when applied to real construction sites.

Application of domain adaptation

To address the domain shift challenge, DA is employed to sustain model performance when transitioning from one data distribution (source domain) to another (target domain). The difference in learning mechanisms between supervised learning and DA is illustrated in Fig. 2, using two different domain datasets. Under supervised learning, a model trained on source data (domain1) may perform poorly on target-domain images (domain2) due to domain shift. By contrast, DA is adopted when the same classification task must be handled in two domains with dissimilar data distributions.

For DA to function effectively, two conditions generally need to be fulfilled. First, the domains should be sufficiently related—if no direct correspondence exists, knowledge transfer is likely to fail[21], and may even degrade performance compared to training without transfer. Finally, both domains should have enough data to produce a robust shared representation; heavily skewed or insufficient datasets in either domain often result in low accuracy in both training and testing[22–25].

Despite these requirements, DA methods can rival or exceed supervised learning while demanding fewer labeled samples. Hong et al.[26] tackled a Single-Sample-Per-Person (SSPP) face detection challenge—using only one passport photo—by employing facial landmark detection to expand the dataset, achieving 97.91% accuracy. Hu et al.[27] bridged the gap between synthetic point clouds from Building Information Modeling (BIM) and real point clouds for semantic segmentation, achieving an average accuracy of 91.03%. Hong et al.[28] tested data from one construction site (source) against three different sites (targets), with DA boosting accuracy from 40.43% to 82.76%. Duan et al.[29] introduced an unsupervised, feature-level DA approach to enhance a Reinforcement

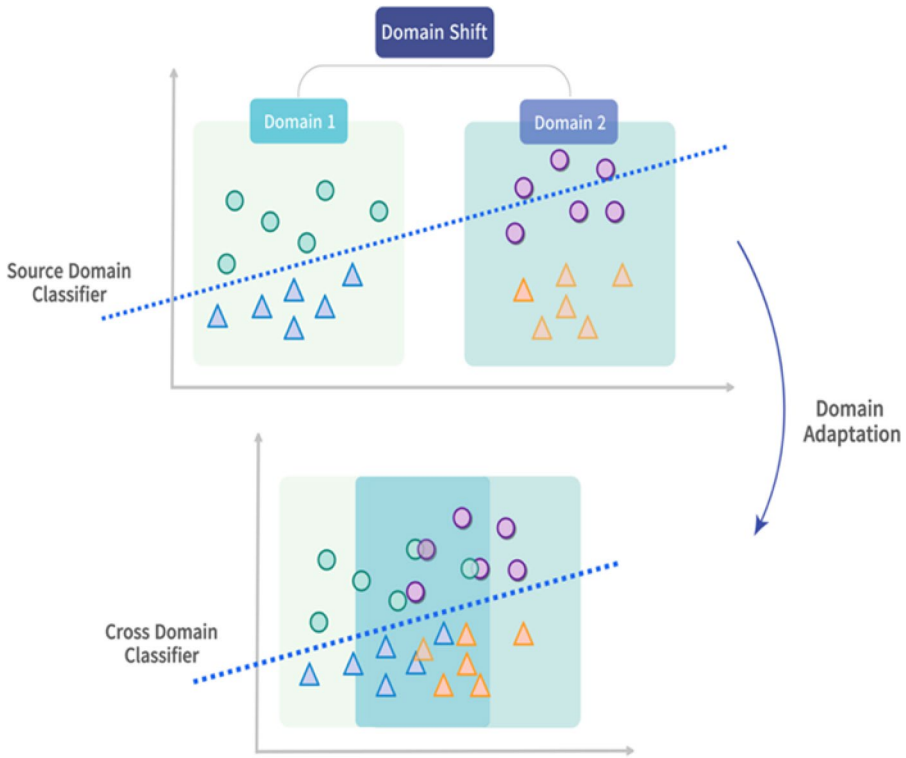


Fig. 2. Visualization of class distributions of source and target domains.

Learning (RL) strategy for cable-in-duct installation, raising a 98% success rate in simulation to 95.8% in real-world conditions. Similarly, Tran et al.[30] demonstrated that adopting DA in worker detection via object detection models at construction sites led to a 93% success rate in real-world applications.

Above studies show that DA can significantly boost performance without relying on large amounts of labeled data, especially in tasks requiring robust transfer across different domains. Building on this, if DA techniques are applied to non-PPE detection using general-context data—combined with carefully designed configurations—high accuracy can be attained. Accordingly, this study introduces a DA-based method specifically tailored for non-PPE detection on construction sites.

Proposed approach

To accurately detect non-PPE, this study proposes a DA–enhanced object-detection framework based on Faster R-CNN. The overall workflow is summarized in Fig. 3. The pipeline begins with two raw input domains: a general-context source domain and a construction-site target domain. Both domains are first passed through an image-augmentation stage, where brightness, scale, perspective, rotation, shearing, and blurring transformations are applied to emulate camera motion, varying viewpoints, and lighting conditions commonly observed on construction sites. The augmented images are then fed into a Faster R-CNN detector, which consists of a shared backbone network, a Region Proposal Network (RPN), and a detection head that predicts bounding boxes and non-PPE categories.

On top of this detector, two adversarial domain-adaptation modules are attached: an image-level adaptation module that operates on the global feature map and an instance-level adaptation module that operates on region-wise features produced by the RPN. Both modules are connected so that the backbone is encouraged to learn domain-invariant features while the detection head is optimized using the standard Faster R-CNN loss. The subsequent subsections describe each component and the overall training and evaluation procedure in detail,

Input domain

The input domain encompasses all images that feed into the processing pipeline, whether originating from the source or target distribution. Although both domains may exhibit similarities (e.g., human features), they can differ substantially in attributes such as background, lighting, and context. The following subsections describe the function of each domain module in greater detail.

Source domain

The source domain module provides a labeled dataset that forms the basis for supervised training. This dataset includes both images and their corresponding labels (e.g., "non-helmet" and "non-vest"), enabling the model to learn how specific visual features map to each category. The knowledge the model acquires—such as recognizing particular textures or color patterns—can then be transferred or adapted to the target domain[31]. Compared to construction-site imagery, the source domain often presents more controlled or diverse backgrounds—whether in offices, on streets, or in general indoor/outdoor settings—leading to notable differences in color palettes, object placement, and subject movement.

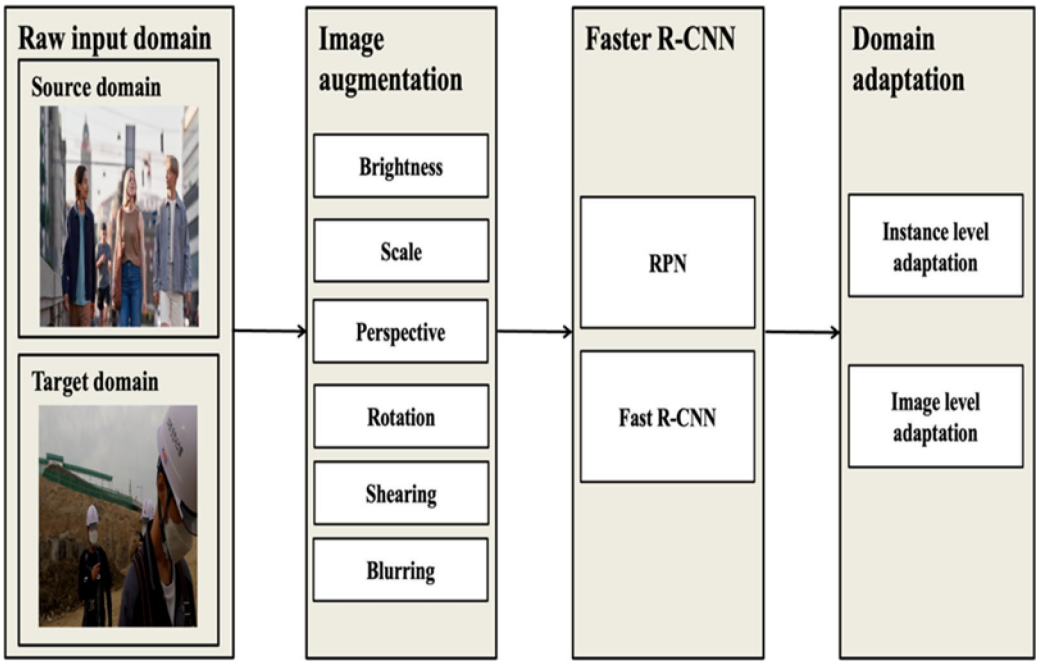


Fig. 3. Overall framework of the proposed method based on DA.

Target domain

In many domain adaptation settings, the target domain consists of images that are sparsely labeled or even entirely unlabeled[32,33]. In this study, however, the target domain corresponds to construction-site images and is fully labeled for all non-PPE categories. The primary motivation for adopting DA here is therefore to address domain shift between a larger, fully labeled general-context source domain and the more limited but fully labeled construction-site target domain, not to solve a labeling problem. Although DA techniques are often used to alleviate annotation burdens in unsupervised or semi-supervised scenarios, in this work they are employed mainly to compensate for contextual and environmental differences (e.g., backgrounds, camera viewpoints, lighting, and motion blur) between the source and target domains.

Image augmentation

On construction sites, cameras typically capture high-resolution images, but these images often include significant challenges such as motion blur, abrupt changes in perspective, and frequent object occlusions. Creating a comprehensive dataset of these images suitable for deep learning applications is highly labor-intensive and time-consuming. To mitigate this issue, image augmentation techniques are applied to artificially increase dataset size. In this research, six specific augmentation methods have been implemented across both source and target domain datasets, each addressing a particular image-quality challenge, as summarized in Table 1. Detailed mathematical descriptions of these augmentation techniques can be found in[34].

Non-PPE images can be collected under diverse indoor and outdoor environments, leading to significant variations in illumination due to factors such as artificial lighting conditions, weather patterns, and time of day. Brightness augmentation addresses these variations by systematically altering overall illumination levels, generating images ranging from dark to brightly lit[35]. Perspective transformation modifies the homography matrix, ensuring consistent recognition of buildings and other elements despite changes in viewpoint angles. Scale augmentation, a type of affine transformation, simulates different object distances and adjusts for varying PPE sizes. Rotation augmentation is particularly helpful for feature detection when the camera captures images at non-horizontal orientations.

When the camera angle is significantly tilted, shear augmentation corrects image distortion by adjusting pixel alignment along predefined axes[36]. Lastly, Gaussian blur emulates the motion blur typically encountered when photographing workers in motion, helping the model robustly detect objects despite blurred conditions. This technique redistributes pixel intensities based on their spatial proximity, smoothing images and enhancing model resilience against common image degradations found in real-world scenarios.

Faster R-CNN

As presented in Fig. 4, Faster R-CNN involves two sub-modules: Region Proposal Network (RPN) and Fast R-CNN. Different functions are carried out simultaneously: (1) RPN attempts to find the potential ROIs of target object. (2) Fast R-CNN is used to carry out the more accurate adjustment of ROIs, and classification of target object's class from potential ROIs generated by RPN[37,38].

There are many deep learning techniques involved in Faster R-CNN as basic components such as Fully Connected (FC) layers, and Stochastic Gradient Descent (SGD). While these techniques are mentioned, it does not explore them in depth. Instead, it focuses on presenting the design of Faster R-CNN. For an in-depth understanding of the detailed description, readers should refer to the specified citation[39].

RPN

The first stage of Faster R-CNN is the RPN, which identifies bounding-box candidates likely to contain objects. The RPN workflow is illustrated in Fig. 5. A deep convolutional backbone, also called the feature extractor, transforms the input image into a high-level feature map. The choice of backbone significantly affects detection accuracy and computational efficiency, as different architectures present various trade-offs in capacity and speed. The selection here includes EfficientNetV2, InceptionV3, ResNet-101, ResNet-152, and MobileNetV3, based on prior research[40,41] demonstrating their effectiveness in Faster R-CNN-related studies and applications.

Once the backbone produces a feature map, the RPN typically includes a 3×3 convolution that slides across this map, followed by two sibling 1×1 convolutional layers for objectness scoring and bounding-box refinement. This arrangement captures local spatial context before generating a set of anchors (typically nine per spatial location) that cover multiple scales and aspect ratios. Two output branches then evaluate each anchor:

Method	Parameters	Value ranges
Brightness	β	[−30, 30]
Scale	α	[0.5, 2.0]
Perspective	x, y	[0.8, 1.2], [0.8, 1.2]
Rotation	$a_{31}, a_{32},$	[0.01, 0.15]
Shearing	θ	[−25°, 25°]
Blurring	θ	[0, 1]

Table 1. Augmentation methods applied, including parameter settings, and value ranges.

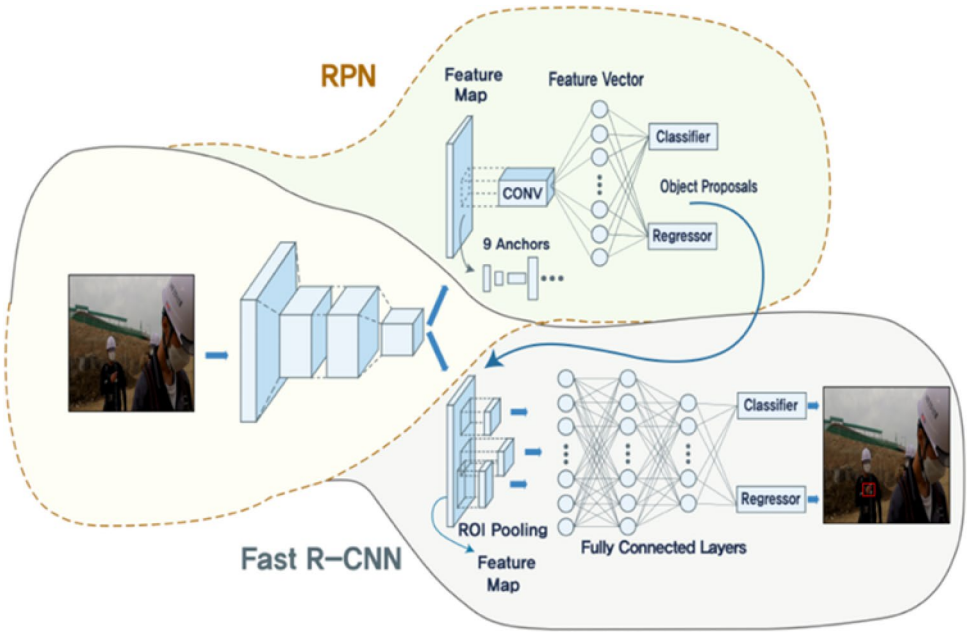


Fig. 4. Workflow of Faster R-CNN.

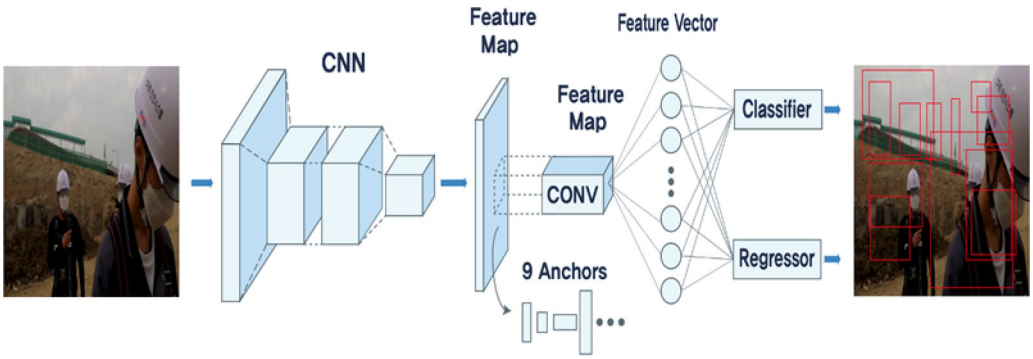


Fig. 5. Workflow of RPN.

- Classifier: Performs binary classification by predicting an objectness score that estimates the likelihood of an anchor corresponding to a foreground object. This branch uses a binary cross-entropy loss, informed by annotations labeling anchors as positive or negative.
- Regressor: Refines anchor coordinates to more accurately approximate object boundaries. A Smooth L1 loss penalizes deviations between predicted coordinates and ground-truth offsets derived from bounding-box annotations.

After generating objectness scores and bounding-box refinements, the RPN ranks anchors by confidence and applies non-maximum suppression (NMS) to remove redundant or overlapping proposals. The resulting set of region proposals is passed to the second stage. In this research, because both the source and target domains are fully labeled, the RPN is trained using bounding-box annotations from both domains.

Fast R-CNN

In the second stage, Fast R-CNN, further refines the object proposals and classifies each one as either one of the known object categories or background. The Fast R-CNN workflow is illustrated in Fig. 6. First, the candidate bounding boxes are mapped onto the CNN backbone's feature map, and ROI pooling aggregates the features within each bounding-box region into a fixed-size feature representation. Subsequent FC layers generate two outputs for each pooled region:

- Classifier: A softmax-based head that yields probabilities for each foreground class plus a background class, indicating whether the proposal corresponds to a specific object category or is background.

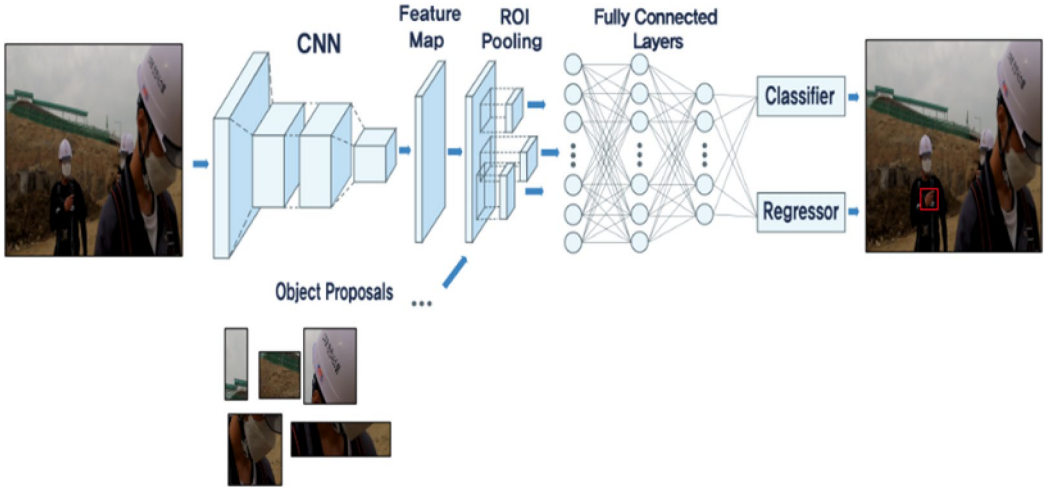


Fig. 6. Workflow of Fast R-CNN.

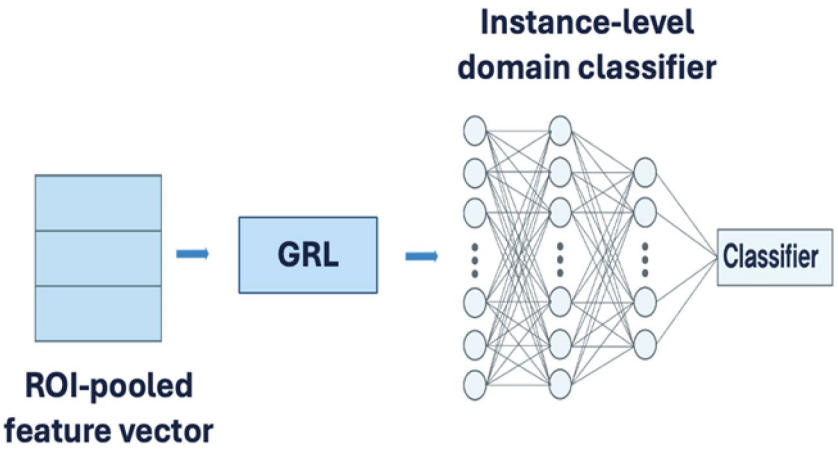


Fig. 7. Workflow of instance-level domain adaptation.

- Regressor: A bounding-box regression head that adjusts each proposal's coordinates to more closely match the underlying object. A Smooth L1 loss quantifies the difference between predicted and ground-truth coordinates.

These outputs form a multi-task detection loss, combining cross-entropy for classification with a bounding-box regression term. In the context of domain adaptation, the intermediate features leading into these heads can be leveraged by an instance-level adaptation module to align object-level representations across domains. Because the target domain is also fully labeled, both source- and target-domain annotations inform this stage during training.

Domain adaptation module

The domain adaptation module aims to mitigate the discrepancy between a labeled source domain and a labeled construction-site target domain by encouraging feature distributions to remain invariant across the two domains, particularly with respect to construction-site–specific factors such as cluttered backgrounds, camera motion, and variations in PPE appearance. The module comprises two primary components: an instance-level module and an image-level adaptation module, both described below.

Instance-level domain adaptation

The workflow of the instance-level domain adaptation module is shown in Fig. 7. This module focuses on localized features extracted by Faster R-CNN, specifically the instance-level representation (i.e., the RoI-pooled feature vector). The RPN first generates bounding-box proposals, each of which is mapped onto the backbone's feature map. A RoI Pooling operation then produces a fixed-size feature vector r_k, capturing the localized appearance cues of the corresponding region.

To align these ROI-pooled features between the source and target domains, an instance-level domain classifier, denoted by $D_\phi^{\text{inst}}(\cdot)$, is employed. This classifier determines whether a given ROI feature vector r_k originates from the source or the target domain. A Gradient Reversal Layer (GRL) precedes, D_ϕ^{inst}, ensuring that, during backpropagation, the gradients are reversed. This mechanism encourages the backbone network to eliminate domain-specific cues from r_k. In this research, D_ϕ^{inst} is implemented using a lightweight FC layer with the following design:

1) Input Layer: Receives the RoI-pooled feature $r_k \in \mathbb{R}^d$.
2) Hidden Layers: Consists of two layers, each comprising 128–256 hidden units and ReLU activations.
3) Output Layer: Contains a single sigmoid neuron, providing the probability that r_k belongs to the source domain.

Let R_S and R_T be the sets of ROI-pooled features derived from source and target images, respectively. Each ROI-pooled feature from the source domain is denoted by $r_k^S \in R_S$, and each ROI-pooled feature from the target domain is denoted by $r_k^T \in R_T$. The adversarial loss, $L_{\text{inst}-\text{adv}}$,encourages the RoI-pooled features to be indistinguishable across domains:

$$L_{\text{inst}-\text{adv}} = -\sum_{r_k^S \in R_S} \log\left[D_\phi^{\text{inst}}\left(r_k^S\right)\right] - \sum_{r_k^T \in R_T} \log\left[1 - D_\phi^{\text{inst}}\left(r_k^T\right)\right] \quad (1)$$

This adversarial objective compels the network to produce RoI-pooled features r_k that are indistinguishable between the source and target domains, thereby mitigating domain shifts at the instance level.

For the construction-site target domain, instance-level alignment is especially important because PPE and non-PPE appearances vary with site, contractor, and season (e.g., different helmet shapes, vest colors, or clothing under vests), and because workers are frequently partially occluded. By aligning RoI-pooled features across domains, the instance-level module focuses the adaptation process on the visual patterns within the proposed bounding boxes, so that non-PPE instances detected in general-context images can support robust recognition of visually different but semantically equivalent non-PPE events on construction sites.

Image-level domain adaptation

The workflow of the image-level domain adaptation module is shown in Fig. 8. While the instance-level module primarily accounts for localized object features, the image-level module addresses global domain discrepancies in the feature map extracted by the CNN backbone in Fast R-CNN. Let $g_k^S = G_\theta\left(\mathrm{x}_k^S\right) and\, g_k^T = G_\theta\left(\mathrm{x}_k^T\right)$ denote the global features extracted by the backbone from the k-th source image x_k^s and k-th target image x_k^T, respectively. Such global features capture large-scale domain differences—such as illumination, color distribution, and background textures—that often diverge substantially between source and target domains. An image-level domain classifier, $D_\phi^{\text{img}}(\cdot)$, is trained to discern whether g_k arises from the source or target domain. As in the instance-level setting, a GRL is placed before D_ϕ^{img}, enforcing the backbone to learn domain-invariant representations at the image level.

After passing through the GRL, a 3 × 3 convolution layer processes the global feature map. This step refines and potentially reduces the dimensionality of the map before feeding it into subsequent layers. The convolved output is then fed into a FC layer to transform the data into a compact feature vector. In this study, the overall architecture of D_ϕ^{img} mirrors that of D_ϕ^{inst}:

1) Input Layer: Receives the global feature vector $g_k \in \mathbb{R}^m$.
2) Hidden Layers: Two hidden layers, each containing 128–256 hidden units and followed by ReLU activations.

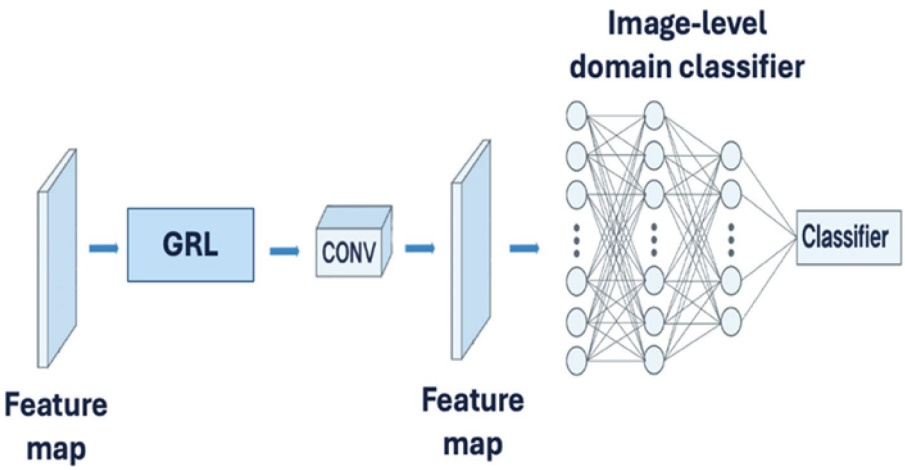


Fig. 8. Workflow of image-level domain adaptation.

3) Output Layer: A single sigmoid neuron that produces the probability of g_k originating from the source domain.

The adversarial loss for global feature alignment, $L_{\mathrm{img-adv}}$, is given by:

$$L_{\mathrm{img-adv}} = -\sum_{\mathrm{x}_k^S \in D_S} \log\left[D_\phi^{\mathrm{img}}\left(g_k^S\right)\right] - \sum_{\mathrm{x}_k^T \in D_T} \log\left[1 - D_\phi^{\mathrm{img}}\left(g_k^T\right)\right] \quad (2)$$

where D_S and D_T are the sets of source and target images, respectively. By minimizing this loss, the image-level features g_k become domain-invariant, thereby reducing global discrepancies between source and target domains.

In contrast, image-level adaptation targets global characteristics of construction sites that differ strongly from general-context scenes, such as large equipment, scaffolding, soil and concrete surfaces, and complex lighting from mixed natural and artificial sources. Without this module, the backbone features tend to encode these background and illumination differences as domain-specific cues, which can reduce detection performance when the camera is moved to a new site. By enforcing alignment of global feature distributions, the image-level module promotes domain-invariant representations of the overall scene, allowing the detector to focus on non-PPE behavior rather than on site-specific environmental context.

Model performance evaluation

AP

In object detection tasks, AP serves as a common metric for assessing detection accuracy. AP is calculated as the average precision achieved across various recall levels, typically spanning from 0 to 1, using different Intersection over Union (IOU) thresholds. IOU is a measure indicating how much overlap exists between a predicted bounding box and its corresponding ground-truth bounding box, relative to the combined area of both boxes[42]. Mathematically, AP can be expressed as follows:

$$\mathrm{AP} = \sum_{i=1}^{n} p\,(i)\,\Delta\, r\,(i) \quad (4)$$

Here, *n* represents the total number of bounding boxes detected, iii refers to the detection ranking, *p(i)* indicates precision calculated up to the *i*-th detection, and Δr(*i*) denotes the incremental change in recall from detection *i* – 1 to i[43].

For non-PPE detection scenarios, using an IOU threshold of 50% (AP50) is particularly beneficial, as it balances rigorous bounding box alignment requirements with practical allowances for minor detection errors. In real-world PPE compliance checks, identifying non-compliance typically does not necessitate pixel-level accuracy; thus, a predicted bounding box overlapping at least half of the ground-truth box is considered sufficient.

Training time, and detection speed

Training duration refers to the total time taken to train a detection model on the provided dataset, serving as an indicator of the computational resources and effort required[44]. Detection speed describes the rate at which a model processes images, typically measured in Frames per Second (FPS). Higher FPS values indicate quicker object classification or detection performance, a crucial factor in scenarios demanding real-time or rapid processing capabilities.

Experiment

Dataset construction

Raw dataset

This study utilized datasets derived from two main sources: on-site data collection and publicly accessible databases. As mentioned earlier, detecting non-PPE use is particularly challenging since real-world imagery generally portrays workers appropriately wearing protective equipment. Nonetheless, recognizing scenarios lacking proper PPE remains essential.

For the on-site data, images were collected at five different construction locations in South Korea. A GoPro HERO9 Black camera was attached to helmets worn by 35 construction workers during peak operational periods. Continuous video recordings were obtained, from which individual frames were systematically extracted to reflect actual working conditions (Fig. 9 (a)). While this process initially produced a substantial collection of 54,329 images, balanced sampling was applied to adequately represent both compliant scenarios (workers fully equipped with PPE) and non-compliant scenarios (instances of missing PPE), resulting in a curated dataset of 1,207 images.

To supplement the on-site data, publicly available datasets were included, specifically the Safety Helmet Wearing Dataset (SHWD)[45], containing 530 images, and the Safety Helmet Computer Vision Project (SHCVP)[46], comprising 565 images. These external resources helped broaden the variety of visual conditions and contexts for effectively detecting non-PPE compliance. Representative examples from these datasets are illustrated in Fig. 9 (b).

By contrast, the general-context dataset comprises images drawn from public streets and everyday settings, where individuals typically do not wear PPE. This subset originates from two public datasets[46,47] which often include images procured via search engines (e.g., Google, Baidu), resulting in scenes that may not reflect real



(a) Images captured by body-worn cameras on construction sites



(b) Images from publicly available construction-site datasets

Fig. 9. Example images of construction datasets.

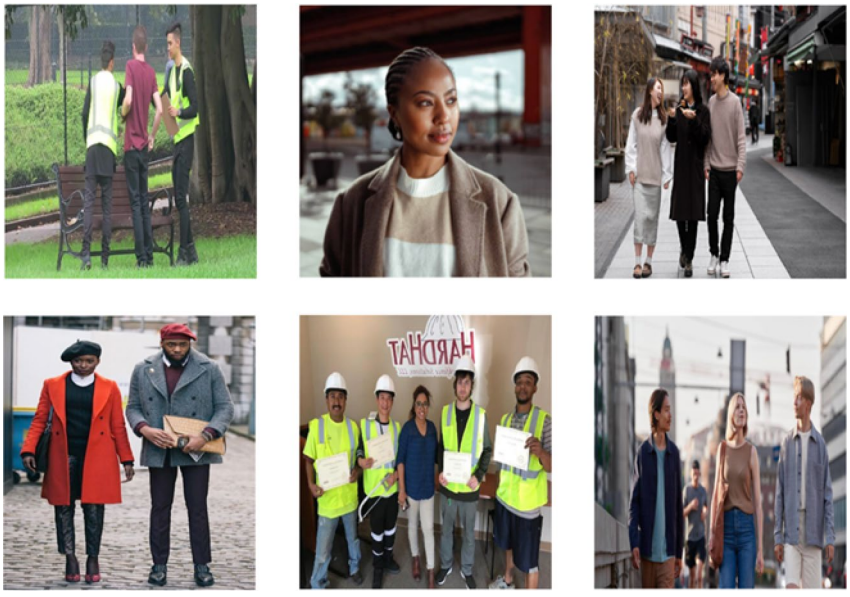


Fig. 10. Example images of general context dataset.

construction environments. A total of 2,174 general-context images were collected. Examples of these images are shown in Fig. 10. To minimize memory usage, all images in both datasets were resized to 1500 × 900 pixels.

Data split

The dataset collected from body-worn cameras, containing 2,302 images, was divided into training, validation, and testing subsets at a ratio of 60:20:20, yielding 1,382 images for training, 460 for validation, and another 460 for testing. In contrast, the general-context dataset comprising 2,174 images was exclusively used for training purposes. This approach aligns with the study's objective to specifically detect non-PPE scenarios within construction environments, rendering the general-context images less relevant for validation and testing.



Fig. 11. Examples of images applying each augmentation technique.

Dataset	Construction context	General context
Original	1,382	2,174
Brightness	1,382	2,174
Scale	1,382	2,174
Perspective	1,382	2,174
Rotation	1,382	2,174
Shearing	1,382	2,174
Blurring	1,382	2,174
Total	8,292	13,044

Table 2. The dataset distribution of augmented datasets.

Implementation of image augmentation

The parameters chosen for image augmentation considerably affect the resulting image quality, thus influencing the performance of the trained models. These augmentation parameters were established through iterative experimentation to produce realistically augmented images, as detailed in Table 1. Geometric transformations—including scaling, rotation, perspective adjustments, and shearing—can shift parts of an image outside their original boundaries, resulting in empty spaces or overlaps. Since deep learning algorithms typically require input images of consistent dimensions[48,49], any resulting empty spaces were filled using white pixels (pixel intensity set to 255). This step ensures uniformity across all augmented images. Visual examples demonstrating the differences between original and augmented images are presented in Fig. 11.

As indicated in Table 2, the original training images—1,382 from the construction-context dataset and 2,174 from the general-context dataset—were expanded through multiple augmentation strategies (including brightness adjustments, scaling, perspective transformations, rotations, shearing, and Gaussian blur). This augmentation resulted in 6,910 additional images for the construction-context dataset (totaling 8,292 images including originals) and 10,870 additional images for the general-context dataset (totaling 13,044 images including originals).

As shown in Table 2, the original training sets—1,382 images from the construction-context dataset and 2,174 images from the general-context dataset—were expanded through various augmentation techniques (brightness, scaling, perspective, rotation, shearing, and blurring). This process produced 6,910 augmented images for the construction-context dataset (resulting in a total of 8,292 images when combined with the originals) and 10,870 augmented images for the general-context dataset (for a total of 13,044 images).

Data annotation

Annotations for non-PPE items were created by manually drawing bounding boxes around regions labeled as "Non-helmet," "Non-mask," "Non-glove," "Non-vest," and "Non-shoes." Any remaining areas were categorized as

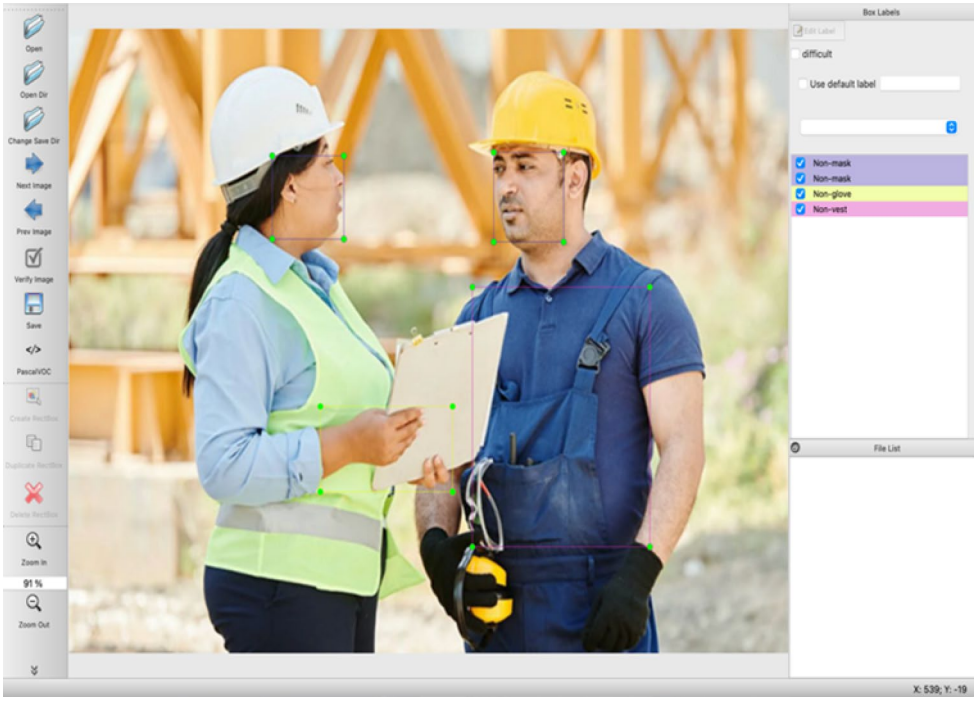


Fig. 12. Examples of annotation process for detecting non-PPE.

Class	Definition	Visual cues
Non-helmet	A worker who is not wearing a protective safety helmet	The head or hair is clearly visible; no helmet shape or color pattern on top of the head
Non-mask	A worker who is not wearing a face mask or respirator	The mouth and nose are uncovered; the lower half of the face is visible without mask material or straps
Non-glove	A worker who is not wearing protective gloves	Hands are fully exposed; no glove or protective covering is present around the hand region
Non-vest	A worker who is not wearing a high-visibility safety vest	Ordinary upper-body clothing with no reflective material or bright colors that typically indicate a safety vest
Non-shoes	A worker who is not wearing safety footwear	Visible footwear that does not meet safety standards, such as slippers, or casual shoes

Table 3. Definition, and visual cues of target non-PPE.

Dataset	Purpose	Number of images	Class					
			Non-helmet	Non-mask	Non-glove	Non-vest	Non-shoes	Total
Construction context	Training	1,382	126	1,082	741	424	547	2,920
	Augmentation	8,292	756	6,492	4,446	2,544	3,282	17,520
	Validation	460	48	352	238	137	182	957
	Test	460	51	337	211	158	177	934
	Total	10,594	981	8,263	5,636	3,263	4,188	22,331
General context	Training	2,174	8,857	8,864	9,912	8,763	8,893	45,289
	Augmentation	13,044	53,142	53,184	59,472	52,578	53,358	271,734
	Total	15,218	61,999	62,048	69,384	61,341	62,251	317,023

Table 4. Detailed breakdown of dataset composition.

"background." The annotation procedure was performed using LabelImg, resulting in labels formatted in VOC XML. An example of the annotation process is shown in Fig. 12. Table 3 details each non-PPE category and explains the visual cues employed for their identification.

Compilation of dataset

To evaluate the effectiveness of the proposed methodologies, datasets from both construction and general contexts were combined and subsequently divided into training, validation, and testing subsets. Each image was systematically annotated with the relevant non-PPE category. Table 4 provides comprehensive details about the annotated dataset, highlighting the distribution of each non-PPE class within each subset.

Training process

Loss function

The proposed training procedure jointly optimizes both the standard Faster R-CNN detection losses and the adversarial losses for domain adaptation. In certain configurations, an additional consistency regularization term, $L_{\text{consistency}}(\theta)$ is introduced to stabilize predictions on unlabeled target images. The combined training objective can be expressed as:

$$\min_{\theta}\left(\left(L_{\text{det}}(\theta)+\lambda_{\text{inst}}\max_{\phi^{\text{inst}}} L_{\text{inst-adv}}\left(\theta, \phi^{\text{inst}}\right)+\lambda_{\text{img}}\max_{\phi^{\text{img}}} L_{\text{img-adv}}\left(\theta, \phi^{\text{img}}\right)+\lambda_{\text{cons}}L_{\text{consistency}}(\theta)\right)\right) \quad (3)$$

where $L_{\text{det}}(\theta)$ denotes the standard Faster R-CNN detection loss, which combines classification (cross-entropy) and regression (Smooth L1) terms. $L_{\text{inst-adv}}$ enforces domain invariance at the ROI (instance) level. $L_{\text{img-adv}}$ aligns the global features across domains. $L_{\text{consistency}}(\theta)$ ensures consistent predictions under small perturbations for unlabeled target images. λ_{inst}, λ_{img}, and λ_{cons} are hyperparameters weighting each respective loss term.

Within this minimax framework, the domain classifiers (D_{ϕ}^{inst}, D_{ϕ}^{img}) try to maximize their ability to distinguish source from target features, while the backbone parameters θ seek to minimize these same objectives through gradient reversal. Consequently, the network learns domain-invariant representations at both object and image levels, ultimately enhancing detection performance—even when source and target domains differ substantially in appearance.

Since both source and target datasets are fully labeled in this study, bounding-box annotations from each domain contribute to the Faster R-CNN detection training. The adversarial modules then provide an additional benefit by explicitly aligning features across domains, thereby reducing the performance gap that can arise from visual discrepancies between source and target environments.

Optimization of the hyperparameters

Although numerous hyperparameters can be tuned to optimize model performance, examining every possible configuration is nearly impossible due to time and computational constraints[6,50]. Consequently, this study focuses on a select subset of hyperparameters. Stochastic Gradient Descent (SGD) serves as the optimizer, using a batch size of either 1 or 2, which yields highly stochastic (per-sample or near-per-sample) weight updates. Within SGD, the current gradient is combined with scaled gradients from previous iterations using a momentum term, which in this study is set to 0.7 or 0.9.

In addition, a weight decay term (L2 regularization) is added to the loss function for all model weights, tested at 0.0005 or 0.001. The learning rate (step size for weight updates) is assigned either 0.00025 or 0.0001. Because GPU memory limits the batch size, it remains 1 or 2, preserving the intensely stochastic nature of updates. Finally, the number of training iterations—which significantly impacts both precision and training time—is set at 30,000 or 50,000. Altogether, these five hyperparameters yield 32 unique configurations, as summarized in Table 5.

Experimental configuration

All experiments in this study were conducted on a Windows 10 (64-bit) system equipped with an Intel® Core™ i9-12900 K processor (16 cores, 5.20 GHz), 64 GB of DDR5 RAM, and an NVIDIA GeForce RTX 3090 GPU with 24 GB of GDDR6X VRAM. The code used in the experiments is available at this repository[51].

Results and discussion

Underfitting and overfitting

In total, 640 models were generated, encompassing five CNN-based architectures, four datasets, and 32 hyperparameter configurations. Due to space constraints, the complete set of training and validation loss curves for these models is not shown here. Instead, Fig. 13 presents representative training and validation losses for models trained using augmented source and target datasets over 50,000 iterations. With the exception of MobileNetV3, whose loss behaves slightly differently after 30,000 iterations, the other architectures demonstrate continued loss reduction beyond 30,000 iterations, followed by a stable pattern. Overall, all models exhibit a steady decline in training and validation losses as training progresses, with minor epoch-to-epoch fluctuations rather than a perfectly smooth descent. This pattern indicates that the models effectively learn from the training data, improving validation performance and converging toward an optimal loss. Furthermore, the consistent drop in training loss across iterations rules out underfitting, while the parallel decrease and close alignment of training and validation losses suggest that overfitting is not a concern.

Effectiveness of hyperparameter on model performance

The impact of various hyperparameters on model performance was assessed using box plots (Fig. 14), which clustered results by each hyperparameter configuration—yielding 20 box plots that captured changes in mAP, resulting from adjustments to batch size, learning rate, weight decay, momentum, and iterations. Table 6 provides a detailed breakdown of these hyperparameter effects across five architectures evaluated under four dataset configurations (Raw Source, RS; Augmented Source, AS; Raw Target, RT; and Augmented Target, AT).

A per-configuration analysis reveals that iteration is often influential in certain architectures (e.g., EfficientNetV2 and ResNet-101), but other hyperparameters— particularly batch size and momentum—can rival or surpass its impact depending on the specific model and data combination. For instance, while iteration yields the highest cumulative gains for EfficientNetV2 (3.70) and ResNet-101 (6.08), other parameters sometimes

Case	Hyperparameters				
	Batch size	Learning rate	Weight decay	Momentum	Iteration
1	1	0.00025	0.0001	0.7	30,000
2	1	0.00025	0.0001	0.9	30,000
3	1	0.00025	0.0005	0.7	30,000
4	1	0.00025	0.0005	0.9	30,000
5	1	0.001	0.0001	0.7	30,000
6	1	0.001	0.0001	0.9	30,000
7	1	0.001	0.0005	0.7	30,000
8	1	0.001	0.0005	0.9	30,000
9	2	0.00025	0.0001	0.7	30,000
10	2	0.00025	0.0001	0.9	30,000
11	2	0.00025	0.0005	0.7	30,000
12	2	0.00025	0.0005	0.9	30,000
13	2	0.001	0.0001	0.7	30,000
14	2	0.001	0.0001	0.9	30,000
15	2	0.001	0.0005	0.7	30,000
16	2	0.001	0.0005	0.9	30,000
17	1	0.00025	0.0001	0.7	50,000
18	1	0.00025	0.0001	0.9	50,000
19	1	0.00025	0.0005	0.7	50,000
20	1	0.00025	0.0005	0.9	50,000
21	1	0.001	0.0001	0.7	50,000
22	1	0.001	0.0001	0.9	50,000
23	1	0.001	0.0005	0.7	50,000
24	1	0.001	0.0005	0.9	50,000
25	2	0.00025	0.0001	0.7	50,000
26	2	0.00025	0.0001	0.9	50,000
27	2	0.00025	0.0005	0.7	50,000
28	2	0.00025	0.0005	0.9	50,000
29	2	0.001	0.0001	0.7	50,000
30	2	0.001	0.0001	0.9	50,000
31	2	0.001	0.0005	0.7	50,000
32	2	0.001	0.0005	0.9	50,000

Table 5. Combinations of hyperparameters.

prevail in specific scenarios (e.g., RS + RT for EfficientNetV2, where learning rate reaches 2.2; AS + RT for ResNet-101, where batch size hits 1.58).

In contrast, InceptionV3 is driven more strongly by momentum overall (3.57), markedly outpacing iteration (1.58) and learning rate (2.75). For ResNet-152 and MobileNetV3, batch size emerges as the most influential hyperparameter (4.95 and 2.36, respectively). Even then, certain dataset settings exhibit notable peaks in learning rate or weight decay (e.g., learning rate = 1.08 in RS + AT for ResNet-152), demonstrating that no single hyperparameter universally dominates.

Analysis of best performing model in each architecture and dataset

Table 7 presents mAP scores for five architectures evaluated under different configurations of raw and augmented source and target datasets. A consistent pattern emerges: models utilizing augmented data in both source and target domains achieve the highest mAP, whereas raw-only setups produce comparatively lower scores.

For example, EfficientNetV2 improves from a raw-only mAP of 67.40% to 84.90% under full augmentation—underscoring the value of adding synthetic samples. ResNet-101 similarly rises from 69.58% to 86.21%, while MobileNetV3 jumps from 61.08% to 77.65%. These increases confirm that comprehensive augmentation across both domains significantly narrows distribution gaps and enhances model robustness.

Examining each class further underscores this improvement. For instance, "Non-helmet" accuracy in EfficientNetV2 climbs from 43.22% to 86.32% under full augmentation, illustrating how synthetic data substantially strengthens representation learning for challenging classes. Comparable gains appear for "Non-mask", "Non-glove", "Non-vest", and "Non-shoes", emphasizing that joint augmentation of source and target datasets yields considerable accuracy boosts across diverse architectures and classes. These findings demonstrate the critical role of exhaustive data augmentation in mitigating domain shifts and enabling robust, generalizable performance in complex real-world contexts.

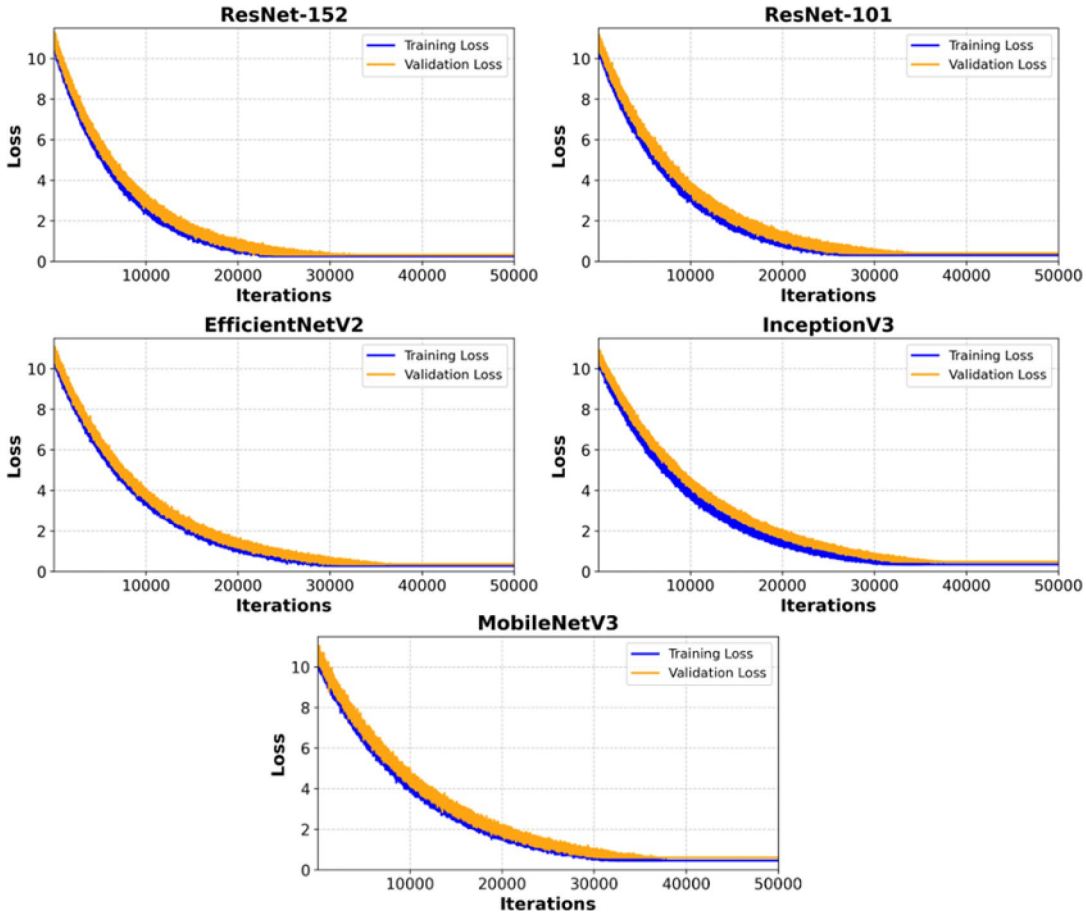


Fig. 13. Training and validation loss graphs.

Tables 8 and 9 summarize the training times for 30,000 and 50,000 iterations and the detection speeds for the five architectures, respectively. At 30,000 iterations, the mean training time is 226.92 min (Std = 2.53), with values ranging from 221.13 to 239.46 min. Increasing the iterations to 50,000 raises the mean training time to 327.96 min (Std = 2.82), spanning from 317.52 to 330.79 min. Regarding detection speed, EfficientNetV2 achieves an average of 27 FPS, while InceptionV3, ResNet-101, and MobileNetV3 each operate at around 23 FPS. ResNet-152 attains the highest detection speed at 33 FPS. For all architectures, the standard deviation of FPS remains below 1 frame per second, indicating highly consistent inference times and demonstrating that the proposed framework can operate within practical real-time constraints.

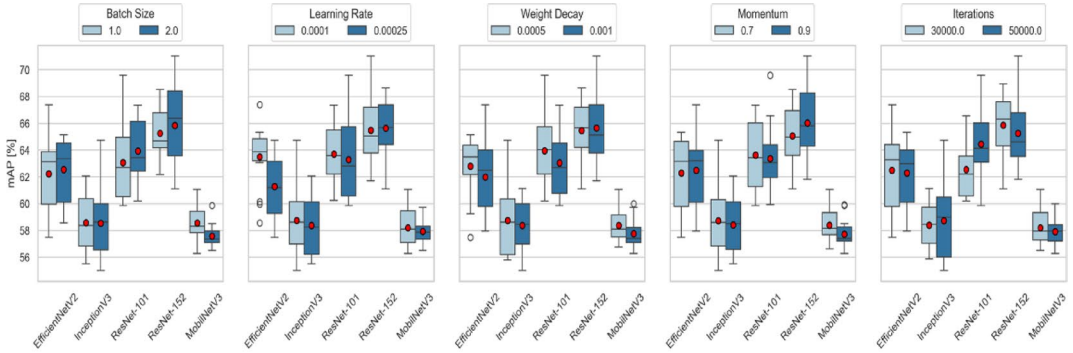
Model test in unseen dataset

Among the various models developed for non-PPE detection on construction sites, the ResNet-152 architecture—trained on both augmented source and target data using hyperparameter configuration Case 27 (batch size = 2, learning rate = 0.00025, weight decay = 0.0005, momentum = 0.7, and 50,000 iterations)—emerged as the top performer. As shown in Fig. 15, its accuracy closely mirrored its validation performance, culminating in an average of 86.84%. These findings suggest that the proposed Faster R-CNN–based domain adaptation framework, combined with image augmentation techniques, holds strong potential for real-world deployment on previously unseen data, contingent on additional evaluation.

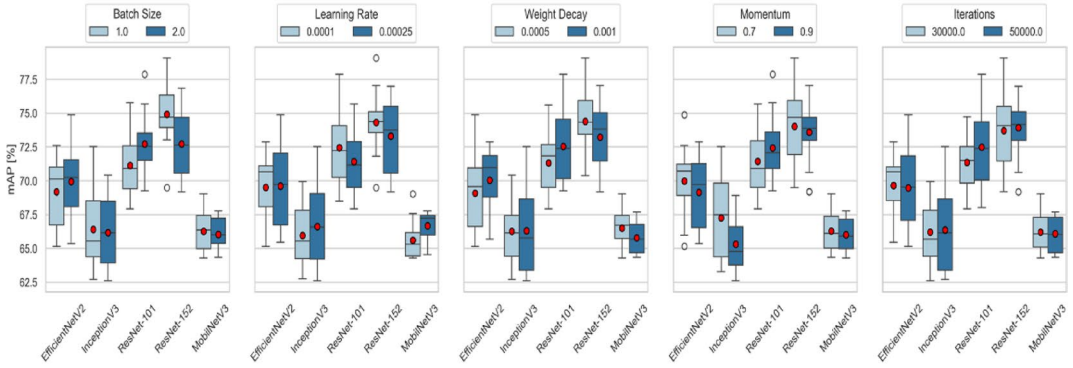
Figure 16 illustrates examples of both correctly and incorrectly detected images. In the correctly detected set (Fig. 16 (a)), workers without safety equipment are accurately identified as "non-PPE." For the incorrectly detected images, errors are highlighted with red circles. In the left image of Fig. 16 (b), the worker on the far left is wearing shoes, yet the detector mistakenly classifies them as "non-shoes." In the right image, a helmet lying on the ground is misidentified as "non-helmet."

Comparative evaluation

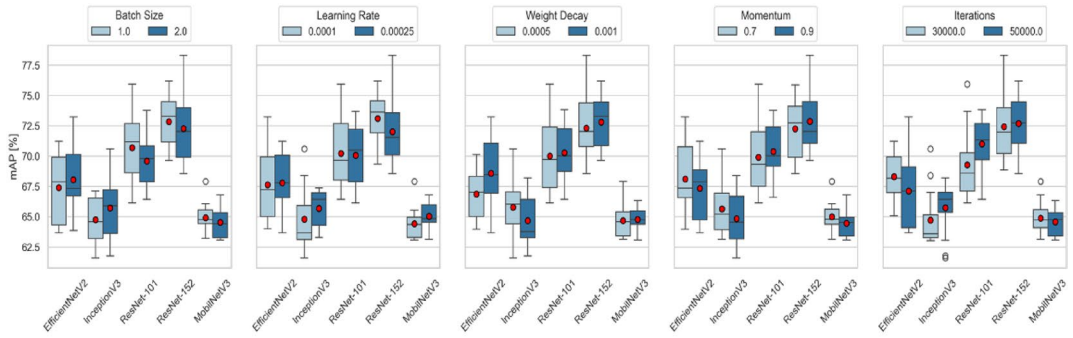
Table 10 summarizes performance across multiple detectors and feature extractors under different source/target data configurations, including standard supervised baselines and the best-performing DA model. Across the supervised baselines (e.g., Faster R-CNN and YOLOv10), mAP generally increases when moving from raw-only training to extensive data augmentation. For example, Faster R-CNN with ResNet-101 improves from 50.38% to 65.41%, while ResNet-152 increases from 47.42% to 72.40% under the same setting. Similarly, YOLOv10 with a ViT backbone improves from 47.60% to 72.21%. Despite these gains, the DA-Faster R-CNN with ResNet-152



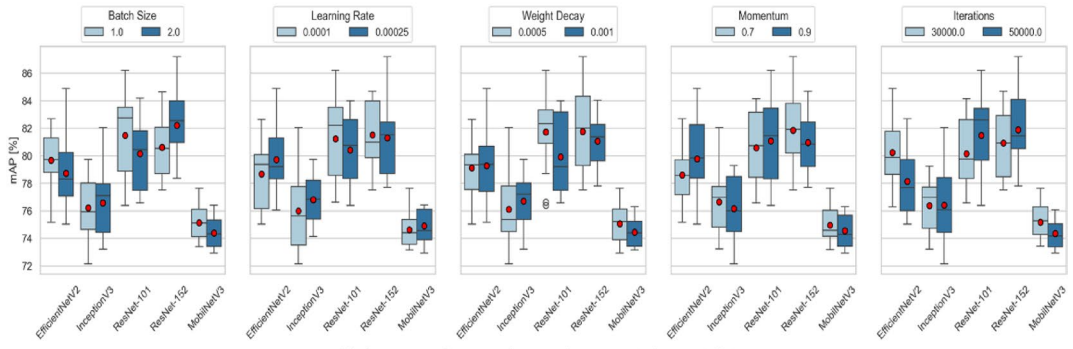
(a) Raw source data and raw target data



(b) Augmented source data and raw target data



(c) Raw source data and augmented target data



(d) Augmented source data and augmented target data

Fig. 14. Impact of each hyperparameters on model performance.

Architecture	Dataset	Hyperparameter				
		Batch size	Learning rate	Weight decay	Momentum	Iteration
EfficientNetV2	RS + RT	0.33	2.2	0.8	0.21	0.21
	AS + RT	0.74	0.09	0.96	0.84	0.19
	RS + AT	0.65	0.15	1.72	0.77	1.2
	AS + AT	0.95	1.04	0.15	1.18	2.1
InceptionV3	RS + RT	0.04	0.38	0.37	0.32	0.35
	AS + RT	0.23	0.65	0.03	1.96	0.16
	RS + AT	0.97	0.9	1.11	0.81	1.03
	AS + AT	0.35	0.82	0.58	0.48	0.04
ResNet-101	RS + RT	0.89	0.42	0.91	0.25	1.9
	AS + RT	1.58	1.04	1.23	0.99	1.12
	RS + AT	1.12	0.15	0.27	0.49	1.73
	AS + AT	1.34	0.83	1.82	0.49	1.33
ResNet-152	RS + RT	0.59	0.16	0.2	0.97	0.61
	AS + RT	2.2	1.01	1.17	0.43	0.21
	RS + AT	0.57	1.08	0.49	0.6	0.24
	AS + AT	1.59	0.22	0.71	0.89	0.97
MobileNetV3	RS + RT	1	0.27	0.61	0.66	0.29
	AS + RT	0.23	1.07	0.71	0.27	0.1
	RS + AT	0.39	0.62	0.1	0.55	0.29
	AS + AT	0.74	0.27	0.62	0.41	0.83

Table 6. Mean variations of mAP by each dataset and architecture.

Architecture	Source data		Target data		Class					
	Raw	Aug	Raw	Aug	Non-helmet	Non-mask	Non-glove	Non-vest	Non-shoes	mAP
EfficientNetV2	◯	-	◯	-	43.22	74.33	66.22	77.32	75.91	67.40
	◯	◯	◯	-	69.78	73.83	75.87	76.56	78.31	74.87
	◯	-	◯	◯	66.16	68.16	71.27	78.93	81.73	73.25
	◯	◯	◯	◯	86.32	86.02	85.21	84.58	82.37	84.90
	Average				66.37	75.59	74.64	79.35	79.58	75.11
InceptionV3	◯	-	◯		41.73	70.94	64.05	74.58	72.45	64.75
	◯	◯	◯	-	67.43	71.35	73.83	75.23	74.81	72.53
	◯	-	◯	◯	63.25	66.58	68.62	76.48	78.12	70.61
	◯	◯	◯	◯	83.19	82.15	84.08	81.56	79.32	82.06
	Average				63.90	72.76	72.65	76.96	76.18	72.49
ResNet-101	◯	-	◯	-	47.25	76.51	68.37	78.45	77.32	69.58
	◯	◯	◯	-	73.65	77.39	78.37	80.56	79.43	77.88
	◯	-	◯	◯	70.45	70.71	73.92	81.48	83.12	75.94
	◯	◯	◯	◯	88.12	85.96	87.81	85.23	83.91	86.21
	Average				69.87	77.64	77.12	81.43	80.95	77.41
ResNet-152	◯	-	◯	-	49.18	72.62	74.91	80.15	78.23	71.02
	◯	◯	◯	-	75.32	76.91	81.07	81.68	80.37	79.07
	◯	-	◯	◯	73.82	72.02	77.12	83.25	85.36	78.31
	◯	◯	◯	◯	**89.23**	**88.94**	**86.89**	**86.19**	**84.73**	**87.20**
	Average				71.89	77.62	80.00	82.82	82.17	78.90
MobileNetV3	◯	-	◯	-	40.12	62.93	63.57	70.56	68.21	61.08
	◯	◯	◯	-	65.31	66.84	69.73	72.45	70.83	69.03
	◯	-	◯	◯	61.25	63.39	66.07	73.25	75.62	67.92
	◯	◯	◯	◯	80.73	80.12	76.68	76.53	74.21	77.65
	Average				61.85	68.32	69.01	73.20	72.22	68.92

Table 7. Model performance by each dataset and architecture.

Iteration	Mean	Std	Min	25%	50%	75%	Max
30,000	226.92	2.53	221.13	221.61	225.15	231.63	239.46
50,000	327.96	2.82	317.52	325.39	327.76	329.32	330.79

Table 8. Training time (minutes) by iterations.

Architecture	Mean	Std	Min	25%	50%	75%	Max
EfficientNetV2	27	0.47	26	27	27	27	28
InceptionV3	23	0.67	22	23	23	23	24
ResNet-101	23	0.47	22	23	23	23	24
ResNet-152	33	0.47	32	33	33	33	34
MobileNetV3	23	0.47	22	23	23	23	24

Table 9. Detection speed (FPS) by architectures.

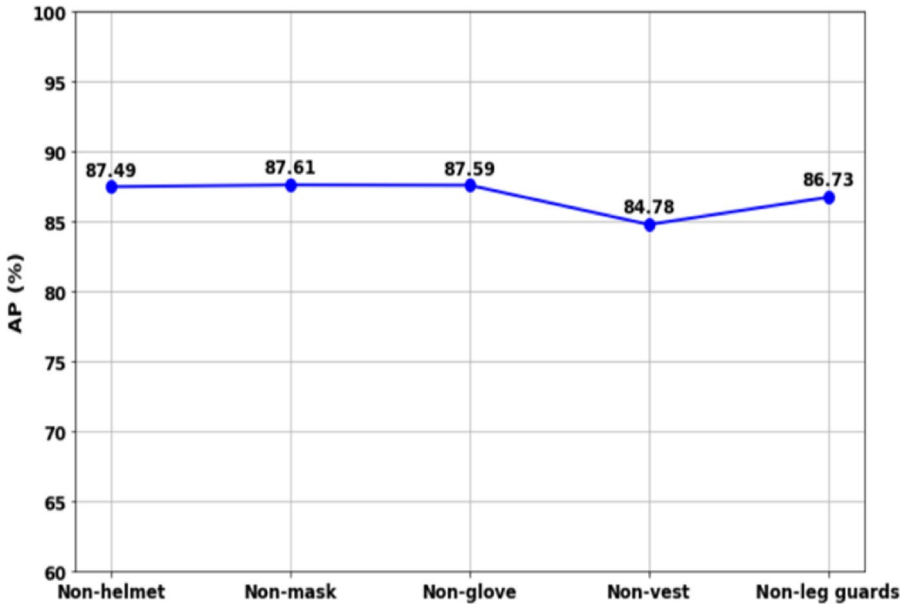


Fig. 15. Best model performance on test data.

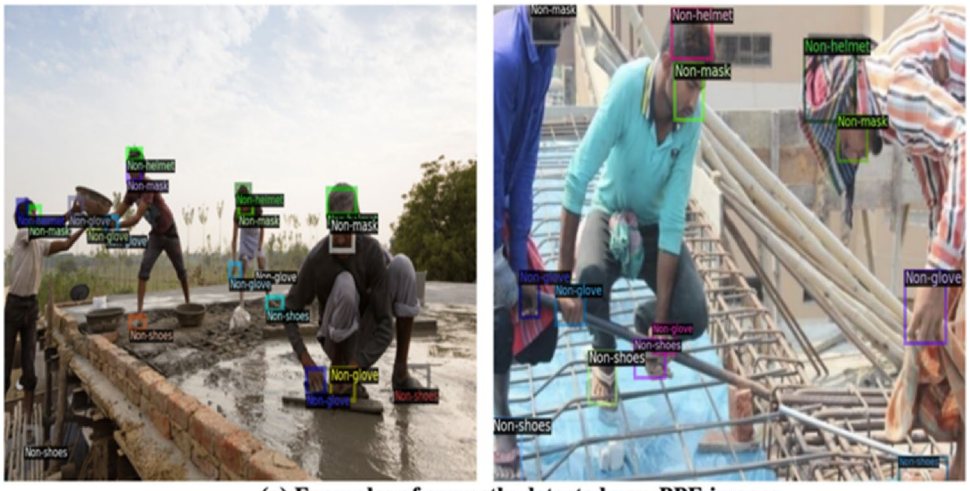
trained with comprehensive augmented data achieves an mAP of 86.84%, outperforming the strongest purely supervised baselines by approximately 14–39 mAP points.

Conclusions

This study presented a DA–based Faster R-CNN framework for non-PPE detection in construction environments, addressing the practical challenge of limited real non-PPE images by leveraging general-context source data. The key findings from the experiments are as follows:

1) Data augmentation in both the source and target domains yields substantial accuracy improvements, increasing mAP by more than 10% points for the DA Faster R-CNN setting.
2) Hyperparameter effects vary by architecture. Although increasing the number of training iterations often leads to noticeable gains, batch size, momentum, and weight decay can be equally influential in some configurations. No clear evidence of severe underfitting or overfitting was observed.
3) Among the five evaluated backbones, ResNet-152 achieved the highest performance (up to 86.84% mAP), outperforming the other CNN- and transformer-based backbones. Despite longer training time, it maintained a practical inference speed of approximately 33 FPS, which is feasible for many on-site applications.

Overall, the results indicate that DA, combined with comprehensive data augmentation, can effectively leverage a limited set of non-PPE examples to build a robust detector. However, this study has several limitations. First, all experiments were conducted using a single construction dataset from one project, which may limit generalizability to other sites and working conditions. Second, the proposed framework was not directly



(a) Examples of correctly detected non-PPE images



(b) Examples of incorrectly detected non-PPE images

Fig. 16. Examples of correct and incorrect results in test data.

compared with other domain-adaptation–capable approaches (e.g., semi-supervised or self-supervised object detection methods), and such comparisons are necessary to better position the method within the broader literature. Third, the dataset was annotated for five non-PPE categories (non-helmet, non-mask, non-glove, non-vest, and non-shoes) and does not include labels for safety ropes, harnesses, or anchoring devices; therefore, it is not directly applicable to detecting non-compliant safety-rope usage, particularly in façade inspection scenarios with distant viewpoints and severe occlusion.

Future work will explore multi-step or continuous DA (e.g., periodic adaptation to new job sites), benchmark the proposed framework against state-of-the-art semi-supervised and self-supervised detectors, and investigate multimodal extensions that combine vision with additional sensor signals (e.g., depth, tension, or vibration) for safety monitoring tasks such as safety-rope compliance. In addition, collecting a dedicated annotated dataset for safety-rope usage and developing lightweight models for edge or embedded deployment are promising directions for real-time safety monitoring in dynamic construction environments.

Source raw	Source aug	Target raw	Target aug	Detector	Feature extractor	Non-helmet	Non-mask	Non-glove	Non-vest	Non-shoes	mAP	Difference
◯	–	◯	–	Faster R-CNN	ResNet-101	33.13	53.34	52.43	57.41	55.59	50.38	−36.46
◯	◯	◯	–			59.53	54.73	54.81	56.65	58.12	56.77	−30.07
◯	–	◯	◯			56.1	56.63	56.78	58.35	61.78	57.93	−28.91
◯	◯	◯	◯			66.26	66.35	67.23	64.39	62.81	65.41	−21.43
◯	–	◯	–		ResNet-152	38.21	48.73	47.65	52.14	50.38	47.42	−39.42
◯	◯	◯	–			59.34	61.19	62.64	65.21	63.09	62.29	−24.55
◯	–	◯	◯			68.01	70.13	69.57	68.44	71.39	69.51	−17.33
◯	◯	◯	◯			71.11	72.02	73.56	72.88	72.44	72.4	−14.44
◯	–	◯	–	YOLOv10	ViT	35.12	47.66	47.89	55.13	52.21	47.6	−39.24
◯	◯	◯	–			58.44	57.82	57.32	61.2	63.17	59.59	−27.25
◯	–	◯	◯			62.66	66.21	67.75	64.72	67.39	65.75	−21.09
◯	◯	◯	◯			70.19	71.05	71.83	73.19	74.81	72.21	−14.63
◯	–	◯	–		Swin Transformer	40.55	52.38	53.12	53.05	54.49	50.72	−36.12
◯	◯	◯	–			60.99	63.53	63.56	65.19	64.1	63.47	−23.37
◯	–	◯	◯			68.43	67.31	68.22	66.12	69.02	67.82	−19.02
◯	◯	◯	◯			72.16	73.88	72.87	72.46	72.11	72.7	−14.14
◯	–	◯	–		MobileViT	38.12	49.99	50.01	50.27	51.56	47.99	−38.85
◯	◯	◯	–			59.92	61.11	61.68	62.53	60.87	61.22	−25.62
◯	–	◯	◯			64.5	66.82	66.17	63.11	65.47	65.21	−21.63
◯	◯	◯	◯			68.71	71.12	70.76	72.23	72.42	71.05	−15.79
◯	–	◯	–	DA-Faster R-CNN	ResNet-152	59.01	76.48	75.88	79.25	77.88	73.7	−13.14
◯	◯	◯	–			68.92	77.65	81.91	81.73	81.12	78.27	−8.57
◯	–	◯	◯			73.71	70.22	82.57	82.99	84.93	78.88	−7.96
◯	◯	◯	◯			87.49	87.61	87.59	84.78	86.73	86.84	0

Table 10. Comparison results of supervised learning and domain adaptation in each dataset.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

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Declarations

Competing interests

The authors declare no competing interests.

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